

Predicate Logic: Predicates, Quantifiers, Free and Bound Variables, Formalization of Statements

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Ac. year 2025-2026

Definition

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If $P(x)$ is a predicate and x has domain D , the **truth set** of $P(x)$ is the set of all elements of D that make $P(x)$ true when they are substituted for x . The truth set of $P(x)$ is denoted

$$\{x \in D : P(x)\}.$$

Definition

The universal quantifier Let $Q(x)$ be a predicate and D the domain of x . A **universal statement** is a statement of the form " $\forall x \in D[Q(x)]$ ". It is defined to be true if, and only if, $Q(x)$ is true for each individual x in D . It is defined to be false if, and only if, $Q(x)$ is false for at least one x in D . A value for x for which $Q(x)$ is false is called a **counterexample** to the universal statement.

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If the context clearly a domain set, then usually write $\forall x$ instead of $\forall x \in D$. For example, in calculus $\forall x$ mean $\forall x \in R$

Truth and Falsity of Universal Statements

- Let $D = \{1, 2, 3, 4, 5\}$, and consider the statement

$$\forall x \in D [x^2 \geq x].$$

Write one way to read this statement out loud, and show that it is true.

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- Consider the statement

$$\forall x \in R [x^2 \geq x]$$

Find a counterexample to show that this statement is false.

The existential quantifier

Definition

Let $Q(x)$ be a predicate and D the domain of x . An **existential statement** is a statement of the form " $\exists x \in D [Q(x)]$ ". It is defined to be true if, and only if, $Q(x)$ is true for at least one x in D . It is false if, and only if, $Q(x)$ is false for all x in D .

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Truth and Falsity of Existential Statements

- Consider the statement

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- Let $D = \{5, 6, 7, 8\}$ and consider the statement

$$\exists m \in D [m^2 = m].$$

Show that this statement is false.

Observe that the two statements “For all real number x , if x is an integer then x is rational”, and “For all integer x , x is rational” mean the same thing because the set of integers is a subset of the set of real numbers. Both have informal translations “All integers are rational”. In fact, a statement of the form

$$\forall x \in U [\text{if } P(x) \text{ then } Q(x)]$$

can always be rewritten in the form

$$\forall x \in D [Q(x)]$$

by narrowing U to be the subset D consisting of all values of the variable x that make $P(x)$ true. Conversely, a statement of the form

$$\forall x \in D [Q(x)]$$

can be rewritten as

$$\forall x [\text{if } x \text{ is in } D \text{ then } Q(x)].$$

Implicit Quantification

Consider the statement

if a number is an integer, then it is a rational number.

As shown earlier, this statement is equivalent to a universal statement. However, it does not contain the telltale word *all* or *every* or *any* or *each*. The only clue to indicate its universal quantification comes from the presence of the indefinite article *a*. This is an example of **implicit** universal quantification.

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Existential quantification can also be **implicit**. For instance, the statement “The number 24 can be written as a sum of two even integers” can be expressed formally as “ $\exists$ even integers m and n such that $24 = n + m$ ”.

Notation

Let $P(x)$ and $Q(x)$ be predicates and suppose the common domain of x is D .

- The notation $P(x) \Rightarrow Q(x)$ means that every element in the truth set of $P(x)$ is in the truth set of $Q(x)$, or, equivalently,

$$\forall x [P(x) \rightarrow Q(x)].$$

- The notation $P(x) \Rightarrow Q(x)$ means that $P(x)$ and $Q(x)$ have identical truth sets, or, equivalently,

$$\forall x [P(x) \leftrightarrow Q(x)].$$

Negations of Quantified Statements

Theorem

The negation of a statement of the form

$$\forall x \in D [Q(x)]$$

is logically equivalent to a statement of the form

$$\exists x \in D [\neg Q(x)]$$

Symbolically,

$$\neg(\forall x \in D [Q(x)]) \iff \exists x \in D [\neg Q(x)].$$

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Interpreting Statements with Two Different Quantifiers

If you want to establish the truth of a statement of the form

$$\forall x \in D \exists y \in E [P(x, y)]$$

your challenge is to allow someone else to pick whatever element x in D they wish and then you must find an element y in E that “works” for that particular x .

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If you want to establish the truth of a statement of the form

$$\exists x \in D \forall y \in E [P(x, y)]$$

your job is to find one particular x in D that will “work” no matter what y in E any one might choose to challenge you with.

Translating from Informal to Formal Language

- “There Is a Smallest Positive Integer” is equivalent to the statement

$$\exists m \in \mathbb{Z}^+ \forall n \in \mathbb{Z}^+ [m \leq n]$$

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- “There Is a Smallest Positive Integer” is equivalent to the statement

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- “There Is No Smallest Positive real Number” is equivalent to the statement

$$\forall x \in \mathbb{R}^+ \exists y \in \mathbb{R}^+ [y < x]$$

- The Definition of Limit of a Sequence:

$$\lim_{n \rightarrow \infty} a_n = L \iff \forall \varepsilon > 0 \exists n_0 \in \mathbb{N} \forall n \in \mathbb{N} [n > n_0 \Rightarrow |a_n - L| < \varepsilon]$$

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If $R(x, y)$ is a predicate about two variables, then

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Example. The sentence

$$\forall x [y \leq x]$$

is predicate which say “any x is greater than or equal y ”. If the domain of sentence is Z^+ , then this predicate is “true” if $y = 0$ and “false” if $y > 0$. If domain is Z , then the predicate is false for all y .

Examples

On the universe $N = \{0, 1, 2, \dots\}$ write the following predicates using the relation “=”, and functions “+” and “.”

- $x = 0$;
- $x = 1$;
- $x = 2$;
- x is even;
- x is odd;
- x is prime.

Definition

An **algebraic system** or a **structure** $\mathcal{M} = (M; \sigma)$ of signature (predicate and functional symbols as $=, + \cdot$ from previous example) σ is defined as a non-empty set M together with an interpretation such that every n -ary predicate (function) symbol from σ is interpreted as an n -ary predicate (function) on M , every individual constant from σ as an element of M .

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Usually the predicate “=” is in all signature and it skips when we write algebraic structures.

Examples

Write down a predicates with two free variables x and y which is true in $\mathcal{M} = (N, +, \cdot)$ if and only if:

- $x \leq y$;
- $x < y$;
- x divides y ;
- x and y are relatively primes.